ITA0518

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CO2-AT3 class test

1. An image appears too bright. Explain and justify the most suitable grey level transformation

A. Technique:

Image Negative Transformation

It covers bright pixels into dark pixels using the formula $S = L - 1 - r$.

This reduces excessive brightness and improves visibility of hidden details.

It is commonly used in medical and photographic image for better visualization.

2. A dark image lacks detail. Describe the suitable grey level transformation.

Technique:

Log Transformation

Formula

$S = c \log(1+r)$

It increases the brightness of low intensity pixels while compressing high-intensity values.

This makes hidden details in dark region clearly visible.

3. An image has poor contrast. explain a histogram based technique.

A. Technique:

Histogram equalization

- It redistributes pixel intensity values over the entire gray-level range.
- this increases image constrat and improves the visibility of objects.
- It is widely used in medical satellite and low constraint images.

4. A low contrast image requires uniform intensity distribution.

A. Technique:

Histogram equalization

- It computes the histogram and redistributes intensity values uniformly.
- the method spreads Grey levels accross the full intensity range.
- The output image has better constrat and improved visual quality

An image shows uneven brightness across regions.

Technique:

Adaptive Histogram Equalization.

- The image is divided into small regions and histograms Equalization is applied for each region.
- It improves local contrast and correets enven illumination.
- It enhances details without deffecting the entire image